

Plexus Atlas — CompuCell3D

The second repository, and the first test of whether any of this transfers

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The twin of `atlas_jax_morph/atlas_note.pdf`. That one asked whether Plexus can say what one framework says. This one asks whether the *procedure* is a procedure.

1. Why a second repository, and why this one

One repository cannot show saturation. The measurement `plexus2.tex` App. E.1 asks for — *does decomposing a broad collection of frameworks converge on a compact reusable vocabulary?* — is a claim about what happens *across* frameworks. With $n = 1$ the curve is a shape, not an argument.

Target: CompuCell3D (Cellular Potts). Nominated by the `jax-morph` note before any of this work, on the grounds that it is *the biggest step away from everything we have*: a cell is not a point with per-cell state, it is a **set of lattice sites**. Its position is a derived centroid. Its dynamics are Metropolis flips of site ownership. There is no per-cell equation to subtract.

That is the point. Every measurement in the `jax-morph` atlas rested on cells being points and a differential test being a subtraction. **If the procedure is general it survives this; if it is secretly a particle-framework procedure, this is where it breaks.**

The prediction — made before running anything, and the reason this note exists — is in `TRANSFER.md`.

2. What transferred, measured rather than asserted

`STATUS.md` in the `jax-morph` atlas claimed ~60% of the Python was instrument and would copy as-is. Forking it is the test of that claim, and it came out slightly better than claimed.

file	edits	why
record.py	none	schema, ladder and the twelve rules are about the record, not the target. Self-test passes unchanged.
saturation.py	none	to <i>transfer</i> : the ledger scores against the frozen baseline and nothing in it knows what a cell is. It was later <i>extended</i> — see §5 — but that is a defect the second repository exposed, not a porting cost.
atlas.py	none	blast radius, rung, commit-or-revert — all HERE-relative.
registry_view.py	none	reads <code>plexus.operators</code> . Produced the same 52-contract baseline from the new folder.
agents/ run_spec.py	none 5 lines	the six roles describe a job, not a repository. output paths only (<code>log/atlas_cc3d/</code> , <code>graphs_data/atlas_cc3d/</code>).
oracle.py, inventory.py, paper.py	rewrite	target-specific by construction: the reference, its mechanism scan, its PDF.

Five of the seven instruments needed zero edits. That is the strongest evidence so far that the apparatus is a method rather than a one-off, and it cost about ten minutes to establish.

3. The oracle — where the risk actually is

TRANSFER.md predicted the oracle would be the largest risk, and a *target* risk rather than a *procedure* risk. It is.

gate	state	
installs	done	CompuCell3D 4.10.0 from its own conda channel into an isolated environment (<code>/workspace/.conda_envs/cc3d-oracle</code>), py312.
imports	done	after supplying <code>libGL</code> (conda-forge <code>libgl-devel</code>) — the container is headless and CompuCell3D links its graphics stack unconditionally.
model definable in Python	done	<code>cc3d.core.PyCoreSpecs</code> : a simulation can be built without XML, which is what makes a scriptable oracle possible at all.
seed exposed	done	<code>PottsCore(random_seed=...)</code> , so determinism is at least <i>expressible</i> .
runs headless	BLOCKED	a bug in 4.10.0, below.
deterministic at a fixed seed	not started	untestable until it runs.

The blocker is in CompuCell3D, not in the plan. Driving it from `PyCoreSpecs` through `service_cc3d` reaches `persistent_globals.get_custom_settings_path()`, which calls `Path(custom_settings_path)` where that value is the *list of specs* rather than a simulation file path:

```
TypeError: argument should be a str or an os.PathLike object ... not 'list'
```

Three routes, in the order they will be tried: an earlier release (4.9.0 / 4.8.0) that may predate the regression; the `.cc3d` project-file route via `cc3d_runScript.sh`, which is the officially supported headless entry point and never touches that code path; or setting the custom-settings path explicitly before the call.

Nothing downstream can be trusted until this gate passes. jax-morph was chosen *because* it ran, and the first note said so in its own Phase 0: a target that cannot be run reduces the atlas to the Okuda situation — wrong operator, wrong parameter or wrong reading of a figure, with no way to tell which.

4. The phases

Seven stages, the same ladder as the first atlas. **I stop at the end of each one and wait for you.** What differs is not the procedure but what each stage has to mean when a cell is a region of a lattice rather than a point.

Phase 0 — Instruments and the oracle

IN PROGRESS

Goal. Make every later measurement trustworthy before taking one.

The instruments came across for free (§2). **The oracle is the phase**, and it is where the risk always was: install, import, run headless, and be deterministic at a fixed seed. Four of those six gates pass; the fifth blocks (§3) and the sixth cannot be tested until it does.

Stop condition: the authors' own model runs twice from one seed and is bit-identical. Until that holds, a later disagreement is unattributable — their noise or our error, with no way to separate them.

Phase 1 — Read every mechanism

not started

The excavator on each mechanism: what it does to the state, its equations, the role of every knob, and the **surprises** a reimplementer gets wrong.

What is different here. jax-morph's mechanisms were 20 classes plus 4 architectural contracts a scan could not see. CompuCell3D's are *plugins* and *steppables* declared through XML or PyCoreSpecs — a different extraction problem, and one where the scan-versus-hand split will fall in a different place. **The 4 hand-added entries were among the most interesting things the first atlas found;** expecting an automatic scan to be sufficient here would be repeating a mistake we already know about.

Phase 2 — Normalize, and let the skeptic attack

not started

Each mechanism becomes a typed Plexus contract with a verdict, then the skeptic tries to break it.

The prediction is already on the record (TRANSFER.md): the *biological* mechanisms — growth, division, death, chemotaxis, secretion, diffusion — should come back largely **alias** or **implementation** of contracts the first atlas already registered, while the *representational* ones — cell-as-domain, the Metropolis acceptance rule, contact energy over shared boundary, volume and surface constraints — should be genuinely new. If that splits as predicted, the conclusion is that *the biological vocabulary is converging while the representational vocabulary is not.*

Phase 3 — Implement what is missing

not started

Every **new** and every **refinement** becomes a Plexus operator in the anti-chamber, with a test of one property statable without the reference: a conservation law, a sign, a limit, a symmetry.

The open question this phase will force. Plexus's cell set is points with per-cell state. A Potts cell is a *set of lattice sites*. Either that is expressible as an existing set with a different map, or the language needs a representation it does not have — and finding out which is worth more than any individual operator here.

Phase 4 — Differential validation

not started

Metric and threshold written down *first*; then the same initial condition through both implementations; then subtract.

This is the phase most likely to break, and the reason this target was chosen. Every differential test in the first atlas was a subtraction of per-cell arrays. Here there is no per-cell equation to compare and the dynamics are a Metropolis chain, so a matched trajectory is not even meaningful — the same argument that made the division differ compare a pooled *hazard* instead of cell counts, but now applying to the whole framework rather than one operator. The tests will have to be **distributional**: equilibrium sorted-boundary length, volume distribution, contact-energy spectrum.

Phase 5 — The forward figure

not started

One headline result of the paper reproduced inside Plexus out of normalized operators, and set *beside* the reference rather than described next to it. Cell sorting is the obvious candidate: it is CompuCell3D's canonical demonstration and its outcome is a topology, not a trajectory.

Phase 6 — Differentiability

not started

Probably not applicable, and saying why is the result. The first atlas's Phase 6 rested on the forward model being differentiable. A Metropolis acceptance is a discrete accept/reject on an energy difference — there is no pathwise gradient through it at all, and the straight-through trick that carried the division draw does not obviously transfer. If Plexus can express Potts dynamics forward but not inversely, that is a real and reportable limit of the language.

Phase 7 — Promotion, and the joint ledger

not started

The measurement only becomes an argument here: the cumulative-new curve across *two* repositories. Flat means the language is saturating; still climbing means it is not.

5. An instrument defect only the second repository could reveal — fixed

`saturation.py` classifies a mechanism **implementation** when it reaches a contract *this run* has already counted as new. At $n = 1$ that is complete. At $n = 2$ it is wrong: every contract the first atlas introduced would be counted as new *again* by the second, inflating precisely the number the instrument exists to protect.

The fix is a `--prior` flag: an earlier `atlas_record.yaml` pre-seeds the "already spoken for" set, so a second sighting reads as an *implementation of that repository* rather than as new vocabulary. Checked by feeding `jax-morph`'s record back to itself — its 8 new contracts all demote and it scores **NEW 0, implementation 14**, which is the correct answer for a repository measured against

itself.

Applied to *both* campaigns' copies, so the instrument stays one instrument. That is the whole value of a second target arriving before the first is promoted: **the defect was invisible at $n = 1$ and structural at $n = 2$** , and it would have silently doubled the headline number.

Where things are. TRANSFER.md — the prediction, written first. ../atlas_jax_morph/STATUS.md — what the first atlas produced and what was reusable. _state/registry_baseline.json — the same frozen 52 contracts, so the two measurements are comparable.